B. K. Birla College of Arts, Science and Commerce, Kalyan (W)

TYBSc Physics, Semester VI Special Theory of Relativity

Assignment 6

Unit 3 (Part B): Introduction to Cosmology and Astronomy in Different Bands

Name: _____ Roll No.: _____ Date: _____

Total marks: 60 **Each question carries 5 marks**

Instructions: Attempt all questions. Section A: theory (write to the point, with neat diagrams where useful). Section B: numerical problems and derivations (state formulae used, show all steps and give units). Sign convention for the interval: $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$; four-vectors $a^\mu = (a^0, \mathbf{a})$ with $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$.

Section A: Theory (6 × 5 = 30 marks)

Q1. Describe Hubble's classification of galaxies with the help of the tuning-fork diagram. Give the main characteristics of elliptical, lenticular, spiral, barred spiral and irregular galaxies. **[5]**

Q2. What are radio galaxies and quasars? Describe the structure of a radio galaxy (core, jets, lobes), the mechanism of its radio emission, and the discovery, properties and energy source of quasars. **[5]**

Q3. Explain how the Doppler shift of spectral lines led to the discovery of the expansion of the Universe. State Hubble's law, explain the meaning of the Hubble constant and the Hubble time, and distinguish the Doppler red shift from the cosmological red shift. **[5]**

Q4. Describe the large-scale structure of the Universe: groups, clusters, superclusters, filaments, walls and voids. How are these structures mapped? State and explain the cosmological principle. **[5]**

Q5. Describe the cosmic microwave background radiation: its prediction and discovery, its spectrum, its origin and its anisotropies. Why is it regarded as strong evidence for the hot Big Bang? **[5]**

Q6. Why is astronomy done in different bands of the electromagnetic spectrum? Compare optical, radio and X-ray astronomy with respect to atmospheric windows, telescopes and typical sources, mentioning interferometry and grazing-incidence mirrors. **[5]**

Section B: Numericals and Derivations (6 × 5 = 30 marks)

Q7. (a) The Hα line (656.3 nm) of a galaxy is observed at 689.1 nm. Find its red shift, its recession speed (using both the classical and the relativistic Doppler formulae) and its distance for $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. (b) An elliptical galaxy has apparent axes $6.0'$ and $2.4'$; find its Hubble type. **[5]**

Q8. Show that Hubble's law $\mathbf{v} = H_0\mathbf{r}$ looks the same to an observer in any galaxy (the expansion has no centre), and that $1/H_0$ estimates the age of the Universe. Calculate the Hubble time for $H_0 = 67.4$ and $73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. **[5]**

Q9. A radio knot in a quasar jet moves with speed v at an angle ϕ to the line of sight. Derive the apparent transverse speed $v_{\mathrm{app}} = v\sin\phi/(1 - \beta\cos\phi)$. Calculate v_{app} for $\beta = 0.98$ and $\phi = 10^\circ$, and find the angle and value of the maximum apparent speed. **[5]**

Q10. Show that the temperature of the cosmic background radiation varies as $T = T_0(1+z)$. For $T_0 = 2.725\,\mathrm{K}$ find (a) the present peak wavelength, (b) the temperature at recombination ($z = 1100$) and at $z = 6$, (c) the present energy density of the radiation. **[5]**

Q11. (a) Find the diffraction-limited angular resolution of the 2.4 m Hubble Space Telescope at 500 nm and of a 305 m radio dish at 21 cm (express in arcsec and arcmin). (b) What baseline would a radio interferometer need at 21 cm to match the resolution of the Hubble telescope? **[5]**

Q12. (a) Using Wien's law, find the peak wavelength and the typical photon energy $k_B T$ for the solar corona ($2 \times 10^6\,\mathrm{K}$) and for the gas in a galaxy cluster ($5 \times 10^7\,\mathrm{K}$). Explain why such observations must be made from space. (b) A quasar varies in brightness over 5 days. Estimate the maximum size of the emitting region in AU ($1\,\mathrm{AU} = 1.496 \times 10^{11}\,\mathrm{m}$). **[5]**

Useful constants: $c = 3.0 \times 10^8$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e c^2 = 0.511$ MeV; $m_p c^2 = 938.3$ MeV; $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m; $1/(4\pi\varepsilon_0) = 9.0 \times 10^9\ \mathrm{N\,m^2/C^2}$; $G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2/kg^2}$; $M_\odot = 1.99 \times 10^{30}$ kg; 1 ly $= 9.46 \times 10^{15}$ m; 1 pc $= 3.086 \times 10^{16}$ m $= 3.26$ ly; 1 yr $= 3.156 \times 10^7$ s; $b_{\mathrm{Wien}} = 2.898 \times 10^{-3}\ \mathrm{m\,K}$; $k_B = 8.62 \times 10^{-5}$ eV/K; $a = 7.566 \times 10^{-16}\ \mathrm{J\,m^{-3}\,K^{-4}}$.